Troubleshooting

Find what you see in the left column. The stages are in order, because a fault in an early stage shows up in every later one.

Uploading and the Serial Monitor

What you see	Likely cause	What to do
No port appears under Tools > Port	The cable only charges, or the board is stuck	Try another cable. Then hold BOOT while plugging in.
"Failed to connect" during upload	The board did not enter upload mode by itself	Hold BOOT, press and release RESET, release BOOT, upload again. Press RESET afterwards.
The Serial Monitor stays empty	Wrong speed, or the wrong board is selected	Set 115200 baud. Select "Nologo ESP32C3 Super Mini". Press RESET with the monitor open.
Garbage characters	Wrong speed	Set 115200 baud.
The light is on when the sketch says off	Your board's light works the other way round	Change `LED_ON` from `LOW` to `HIGH`.

The sensor, stages 2 to 4

What you see	Likely cause	What to do
Stage 2 finds nothing	A wire is open or two are swapped	Check each wire with the multimeter. SDA and SCL are the usual pair.
Still nothing	The sensor is waiting in its other mode	Connect the sensor's `CS` pin to `3V3`.
"This looks like an MPU6050"	The board is a different sensor	This firmware needs a BMI160. See make-changes.pdf.
The address is 0x68 one time and 0x69 the next	`SA0` is not connected	Join `SA0` to `GND`.
Stage 3: "did not wake up"	Poor power to the sensor	Check `3V3` to `VIN` and `GND` to `GND`. Measure 3.3 volts across them.
Stage 3 fails the gravity check	The board moved, or a joint is poor	Send `r` with the board untouched. Reflow the joints of SDA and SCL.
Readings stop now and then	A loose wire	Press on each joint while watching the monitor.
Speed is not 0 when the board is still	The still position was measured while it moved	Send `r`, or in stage 8 hold BOOT for a second, with the board still.
Stage 4 says the gyroscope hit its limit	The swing was faster than 2000 degrees per second	Nothing is wrong. That is the top of the sensor's range.

Bluetooth, stages 5 and 8

What you see	Likely cause	What to do
The test page says the browser blocks Bluetooth	The page was opened as a file	Serve it from `localhost`. The README of stage 5 has the command.
No device appears in the list	The sketch is not running, or something else is connected	Look for the slow blink. Close other tabs and phone apps that use the bat. Switch the computer's Bluetooth off and on.
It connects and nothing arrives	Notifications did not start	Disconnect, press RESET on the board, connect again.
Fewer than 40 readings a second	Weak signal, or a busy computer	Move closer. Close other Bluetooth devices.
The range is only a metre or two	Some Super Mini boards have a weak antenna	Keep the bat near the computer or phone. Another board may do better.
The dashboard connects and shows nothing	It only reads while a shot is being recorded	Press START SHOT.
leg and off side are swapped, or speed is low on a real swing	The board sits in a different direction than the dashboard expects	See "Mount it on the bat" in the README of stage 8.

The battery, stages 6 and 7

What you see	Likely cause	What to do
Nothing happens when switched on	Switch wired to the wrong pins, diode the wrong way round, or the charger's protection has cut off	Measure, with the switch on: charger `OUT+` to `OUT−`, then the `5V` pin to `GND`. The first should read the cell voltage, the second about 0.3 volts less.
`OUT+` to `OUT−` reads 0 with a charged cell	The protection has tripped	Plug the charger into USB for a few seconds. That resets it.
Stage 6 counts brownouts	The path from the cell is too weak	Shorten and thicken the wires. Reflow the joints. Add a 220 µF capacitor from `5V` to `GND`, plus lead to `5V`.
It restarts during a hard swing	A joint opens for a moment	The same. Also fix the cell and the board so they cannot move against each other.
The charger gets warm	It is charging at 1 amp	Normal. It should not be too hot to touch.
The charger's light never changes to "full"	The bat is on while charging	Switch off while charging.
Stage 7 reads "no battery seen"	The switch is off, or R1 is not joined to the switch output	Switch on. Measure the middle point of R1 and R2: it should be half the cell voltage.
Stage 7 reads a little too high or low	The resistors are not exactly equal	Set `CALIBRATION` as the stage describes.

What you see	Likely cause	What to do
The bat blinks five times and goes dark	The battery is empty	Charge it, then switch off and on.

WiFi, stage 9

What you see	Likely cause	What to do
It never joins the network	Wrong name or password, or a 5 GHz network	Check both. The chip only joins 2.4 GHz networks.
"The API was not reached"	Wrong address, a firewall, or different networks	Check the address in `API_URL`. Allow port 5050. Put the computer on the same WiFi.

When nothing explains it

Go back to the last stage that worked and upload its sketch again. If that stage still works, the fault is in what you added since.